

Developer Manual

Metal Spinning CNC Controller

Documentation

Developer Manual - Metal Spinning CNC Controller

For Programmers / System Integrators

Quick Reference

Task	File to Edit
Add error code	`02_DataBlocks.scl` + `06
Change limits	`02_DataBlocks.scl` DB_Ma
Modify motion	`03_AxisControl.scl`
Add G-code command	`05_GcodeHandler.scl`
Add main state	`06_MainProcess.scl` FB_P

Architecture Overview

[Image not found: img_architecture.png]

Key principle: Lower files don't depend on higher files. You can modify 03 without touching 01-02.

Adding a New Feature

Example: Add M3/M5 Spindle to G-code

Step 1: Add outputs to 05_GcodeHandler.scl FB_GcodeParserText:

```
VAR_OUTPUT
    SpindleStartReq : Bool; // Add this
    SpindleStopReq  : Bool; // Add this
END_VAR
```

Step 2: Add parsing in ST_PARSE state:

```
#pM := FIND(IN1 := #sLine, IN2 := 'M3');
IF #pM > 0 THEN
    #SpindleStartReq := TRUE;
    #state := #ST_NEXT;
    RETURN;
END_IF;
```

Step 3: Handle in 06_MainProcess.scl FB_Process.

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Example: Add New Error Code

Step 1: Add to 02_DataBlocks.scl DB_ErrorCodes:

```
ERR_MY_ERROR : Word := 16#0601;
```

Step 2: Add text in 06_MainProcess.scl FB_AlarmManager:

```
16#0601: #ActiveErrorText := 'My Error Description';
```

Code Conventions

Convention	Example
State constants	`ST_IDLE`, `ST_RUNNING`
State values	0=idle, 99=done, 999=erro
FB instance prefix	`fb` -> `fbMoveX`
Bool flag prefix	`b` -> `bTrigMove`
Error codes	Hex 0xAABB (AA=category)

Testing Checklist

- ☐ Compile with no errors
- ☐ Test in simulation
- ☐ Verify E-Stop works
- ☐ Test soft limits
- ☐ Test pause/resume
- ☐ Test error recovery
- ☐ Verify HMI tags work

Debugging Tips

Motion Not Working

1. Check DB_Diagnostics.DrivesReady
2. Check DB_Diagnostics.SafeToRun
3. Check DB_Diagnostics.LimitError
4. Verify TO configuration

G-code Not Executing

1. Check DB_GcodeText.TotalLines > 0
2. Check DB_HMI.UseTextParser = TRUE
3. Check DB_HMI.CurrentLine
4. Check DB_HMI.ErrorID

Modification Rules

If you modify...	Check also...
UDT_RecipeData	All Recipe DBs
DB_HMI	FB_Process, HMI project
DB_MachineConfig	G-code handlers
State constants	Nothing (local)
Error codes	FB_AlarmManager